vitamin **precursor**), frost- and salt-tolerant tomatoes, delayed-ripening pineapples and bananas, canola with a healthier oil profile, and cotton and trees altered to make it easier to process fabric and paper. Some transgenic combinations are strange. Macintosh apples that have been given a gene from a *Cecropia* moth that encodes an antimicrobial protein, for example, are resistant to a bacterial infection called fire blight.

precursor a substance from which another is made

Regulatory Concerns

Whether a new variety of crop plant presents a hazard to human health depends upon the nature of the trait, not how the plant received that trait. For example, the U.S. Department of Agriculture found that a variety of potato obtained through conventional breeding was very toxic, and so it was never developed as a food. However, a potato developed through genetic modification at about the same time did not contain the toxin and was apparently safe to eat. This is why U.S. government regulatory agencies do not evaluate crops on how they were developed, but on their effects on the digestive tracts of animals.

Even after government agencies approve the marketing of a GM crop, consumer acceptance is crucial to its success. The FlavrSavr tomato, for example, was introduced in the 1980s. It ripened later, while in the supermarket, which extended its shelf life while providing an attractive product. However, the developers had focused only on this characteristic, and the tomatoes just did not taste very good. Consumer objection to GM foods also contributed to the FlavrSavr's failure. However, a high-solids GM tomato sold in England before the anti-GM movement began was popular with consumers, largely because it was priced lower than other tomatoes.

The Technique of Genetic Modification

The first step in developing a transgenic plant is to identify a trait in one type of organism that would make a useful characteristic if transferred to the experimental plant. The components of an experiment to create a transgenic plant are the gene of interest, a piece of "vector" DNA that delivers the gene of interest, and a recipient plant cell. Donor genes are often derived from bacteria, and are chosen because they are expected to confer a useful characteristic, such as resistance to a pest or pesticide.

To begin, the donor DNA and vector DNA are cut with the same **restriction enzyme**. This creates hanging ends that are the same sequence on both of the DNA molecules. Some of the pieces of donor DNA are then joined with vector DNA, forming a recombinant DNA molecule. The vector then introduces the donor DNA into the recipient plant cell, and a new plant is grown.

restriction enzyme an enzyme that cuts DNA at a particular sequence

For plants that have two seed leaves (dicots), a naturally occurring ring of DNA called a Ti plasmid is a commonly used vector. Dicots include sunflowers, tomatoes, cucumbers, squash, beans, tomatoes, potatoes, beets, and soybeans. For monocots, which have one seed leaf, Ti plasmids do not work as gene vectors. Instead, donor DNA is usually delivered as part of a disabled virus, or sent in with a jolt of electricity (electroporation) or with a "gene gun" (particle bombardment). The monocots include the major cereals (corn, wheat, rice, oats, millet, barley, and sorghum).

organelle membrane-bound cell compartment

Transgenesis in plants is technically challenging because the transgene must penetrate the tough cell walls, which are not present in animal cells. Instead of modifying plant genes in the nucleus, a method called transplastomics alters genes in the chloroplast, which is a type of **organelle** called a plastid. Chloroplasts house the biochemical reactions of photosynthesis. Transplastomics can give high yields of protein products, because cells have many chloroplasts, compared to one nucleus. Another advantage is that altered chloroplast genes are not released in pollen, and therefore cannot fertilize unaltered plants. However, it is difficult to deliver genes into chloroplasts, and expression of the trait is usually limited to leaves. This is obviously not very helpful in a plant whose fruits or tubers are eaten. The technique may be more valuable for introducing resistances than enhancing food qualities. Someday, transplastomics may be used to create "medicinal fruits" or edible vaccines.

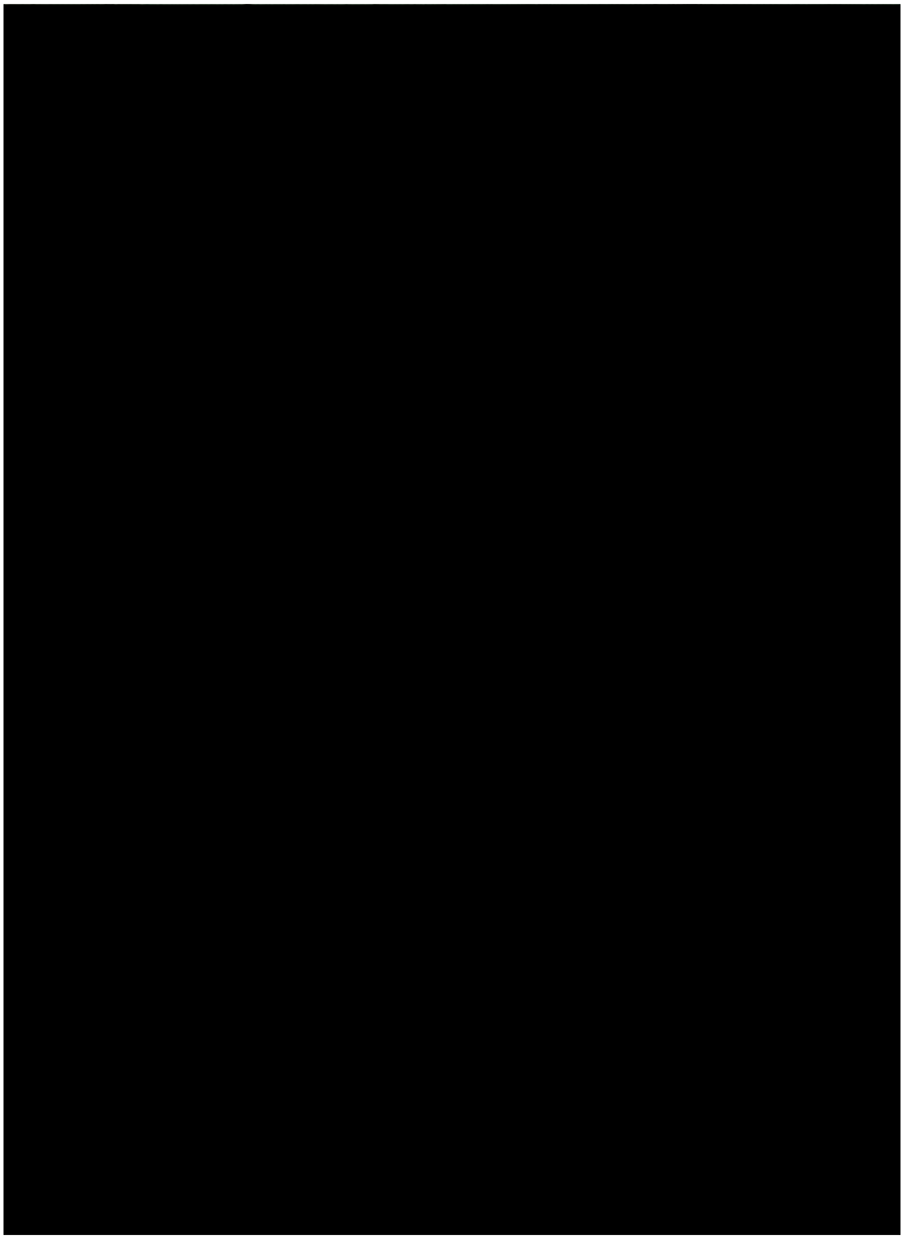
GM beyond the Laboratory

After genetic modification, the valuable trait must be bred into an agricultural variety. Consider "golden rice," a grain that was given genes from daffodils and a bacterium to confer on it the ability to manufacture beta-carotene, a precursor to vitamin A. The first golden rice plants were created solely to show that the manipulation worked, and the modification of an entire biochemical pathway took a decade. The plant varieties were not edible, and the production of beta-carotene was low. In early 2002, however, researchers at the International Rice Research Institute in the Philippines began using conventional breeding to transfer the ability to produce beta-carotene from the inedible golden rice into edible varieties.

Genetic manipulation of plants can also focus on a particular species' own genes. This is the case for the potato, which has traditionally been difficult to cultivate because edible varieties must have an acceptable taste and texture, yet lack the alkaloid toxins that many natural strains produce. Breeding for so many characteristics is very time-consuming, and this is where genetic manipulation might speed the process. Researchers have identified a group of disease resistance genes on a region of one potato chromosome. The genes provide resistances to various insects, nematode worms, viruses, and *Phytophthora infestans*, which caused the blight infection that resulted in the nineteenth-century Irish potato famine. Being able to manipulate and transfer these genes will help researchers quickly breed safe and tasty new potato varieties, and perhaps transfer the potato's valuable resistance genes to related plants, such as tomatoes, peppers, and eggplants.

GM crops are widely grown in some countries, but are boycotted in others where many people object to genetic manipulation. As of 2001, 75 percent of all food crops grown in the United States were genetically modified, including 80 percent of soybeans, 68 percent of cotton, and 26 percent of corn crops. Farmers find that GM crops are cheaper to grow because their reliance on pesticides and fertilizer is less and a uniform crop is easier to harvest. Heavy reliance on the same varieties may be dangerous, however, if an environmental condition or disease should arise that targets the variety, but this dilemma also arises in traditional agriculture.

Because GM crop use is so pervasive in the United States, and because regulatory agencies evaluate the chemical composition and biological effects



This researcher inspects a papaya plant that scientists have attempted to modify to make it resistant to a destructive virus. The benefits and drawbacks of developing, growing, and selling genetically modified crops have been a matter for public debate for decades.

of crops rather than their origin, a consumer would not know that a fruit or vegetable has been genetically modified unless it is so labeled. Some people argue that these practices prevent consumers from having a choice of whether or not to use a genetically modified food. SEE ALSO Agricultural Biotechnology; Antisense Nucleotides; Plant Genetic Engineer; Prion; Restriction Enzymes; Transgenic Organisms: Ethical Issues; Transgenic Plants.

Ricki Lewis

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Geneticist

Genetics intersects almost every other field of biology. For this reason, professionals with a genetics education have a broad range of career opportunities. The recent success of the Human Genome Project has created a great demand for genetics professionals with a variety of expertise in all areas of genetics, not only those applicable to human disease.

Geneticists are involved in identifying genes responsible for basic biological traits, including genes that cause disease or mediate a response to a medication. Geneticists also determine how those genes function in organisms, including plants, animals, and humans. They may treat individuals with a particular genetic disorder or counsel families regarding the genetics of human disease. Geneticists are also involved in identifying genetic mechanisms to improve agricultural processes, such as breeding pest-resistant fruits and vegetables. Geneticists are also interested in the distribution of genetic variations in populations and how those variations arise. Additional job responsibilities may include educating students and the general public, lobbying Congress to pass bills to support genetic research, and testifying in legal cases on the probability that a suspect committed a crime, based on evidence from DNA forensics.

Because the responsibilities of geneticists are highly varied, the educational requirements of geneticists are also quite varied. Individuals with a bachelor's degree are qualified to perform many laboratory procedures, but most often individuals in the field of genetics will obtain a graduate degree. Those geneticists with master's degrees include highly skilled laboratory geneticists and genetic counselors. Geneticists who direct their own research projects must have a Ph.D. in genetics or an M.D. with specialty training in genetics.

Geneticists with a Ph.D. are highly trained professionals who perform genetic research in the areas of expertise mentioned above. Individuals with an M.D. are uniquely skilled in the treatment of patients with genetic disorders and are also often involved in research projects. Many individuals combine a genetics education with learning in other subjects, such as business, law, or computer science. Once having obtained the proper education, geneticists will find job opportunities at academic centers (such as colleges and universities), in government agencies (such as the National Institutes of Health, the Food and Drug Administration, and the Federal Bureau of Investigation), and in industry (for example, pharmacogenetics companies). The salaries paid to genetics professionals will range from $25,000 to well over $100,000, depending on the level of education, area of educational expertise, and job environment (academic, federal, or industrial).

A career in genetics can be exciting; and this is true for all areas of education, expertise, and job environment. It is exhilarating to have the opportunity to participate in research that impacts the treatment of a disease, or improves the crop yield for farmers, or helps correctly identify individuals who commit crimes. SEE ALSO CLINICAL GENETICIST; DNA PROFILING;

Genetic Counselor; Genetics; Molecular Biologist; Pharmacogenetics and Pharmacogenomics; Plant Genetic Engineer; Population Genetics; Statistical Geneticist.

Allison Ashley-Koch

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Genetics

Genetics is the scientific study of the structure, function, and transmission of genes in living things. The field of genetics includes many disciplines and uses many different techniques. Historically, genetic scientists (geneticists) investigated patterns of inheritance in whole organisms by observing the distribution and segregation of physical characteristics across several generations of breeding. This type of genetic research, called classical or Mendelian genetics, is still conducted today and remains invaluable for identifying and describing inheritance patterns and traits. The development of modern molecular biology and biochemistry has facilitated the growth of a different but complementary branch of genetic research, known as molecular genetics. This branch focuses on understanding the detailed molecular mechanisms that govern the transmission of genetic material from one generation to the next.

The observable characteristics that describe any organism (for example, height or eye color in humans, or flower size and color in plants) can be broadly grouped into two categories: those that are acquired because of environmental effects, and those that are inherited. Genetics is concerned with this second category—that is, inherited characteristics. Gregor Mendel, an Augustinian monk of the mid-nineteenth century, observed that characteristics such as shape and color in peas were passed from parent to offspring regardless of the environment the plants grew in. He meticulously counted and documented thousands of crosses between different pea varieties to deduce the principles that governed this inheritance. In the end, his work was so influential and vital to the development of genetics that the term "Mendelian" genetics is now a synonym for classical genetics.

Several key concepts put forward by Mendel have been expanded, as the science of genetics has grown. It is now known that genetic information is passed on as a series of discrete units known as genes, each of which is associated with specific traits. Furthermore, most organisms (including humans) get two copies of their genetic information, one from each parent. This means that most living things have two copies of each gene, and that these two copies are not necessarily the same, since they came from different parents. When an organism reproduces, it passes only one of its two copies to an offspring. Importantly, copies of different genes separate (segregate) randomly into the next generation, which means that an offspring can receive either of the two copies that the parent has, and that the set of copies that is passed is different from offspring to offspring.

A growing interest in the biochemical basis of life led geneticists of the twentieth century to attempt to identify the substance that carries genetic information from one generation to the next. Intense research led to the discovery of DNA (deoxyribonucleic acid), the molecule that encodes the genetic information of almost all organisms (a few viruses use RNA—ribonucleic acid—to encode their genes). Understanding the relationship between observable, inherited traits and the structure and organization of the genetic material (DNA) is the primary focus of molecular genetic research. The powerful combination of classical genetics and molecular genetics has led to many important advances in biological and health-care research and continues to be a major force in the advancement of science in the twenty-first century. SEE ALSO Gene; Geneticist; Genome; Inheritance Patterns; Mendelian Genetics; Molecular Biologist.

Daniel J. Tomso

Genome

A genome is the complete collection of hereditary information for an individual organism. In cellular life forms, the hereditary information exists as DNA. There are two fundamentally distinct types of cells in the living world, prokaryotic and eukaryotic, and the organization of genomes differs in these two types of cells.

Prokaryotes comprise the bacteria and archaea. The latter were originally designated "extremophiles" because they favor such extreme environments as high acidity, **salinity**, or temperature. Prokaryotic cells tend to be very small, have few or no cytoplasmic organelles, and have the cellular DNA arranged in a "nucleoid region" that is not separated from the remainder of the cell by any membrane. Eukaryotes exist as unicellular or multicellular organisms. Among the unicellular eukaryotes are the protozoa, some types of algae, and a few forms of fungi, while the multicellular organisms include animals, plants, and most fungi.

salinity of, or relating to, salt

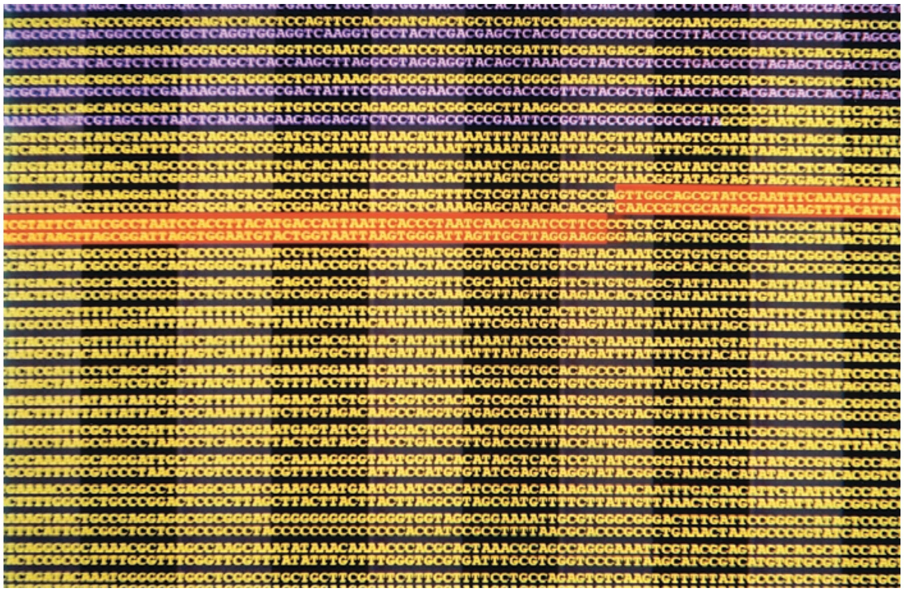
Eukaryotic cells are larger than prokaryotic cells, have a complex array of cytoplasmic structures, and have a prominent nucleus that communicates with components in the cytoplasm through an elaborate nuclear envelope. The hereditary information occurs principally in the nucleus of eukaryotic cells; in addition, minuscule (but essential) amounts of hereditary information occur in some cytoplasmic organelles (specifically, in **chloroplasts** for plants and algae, and in **mitochondria** for all eukaryotic groups).

chloroplasts energy-producing cell organelle

mitochondria the photosynthetic organelles of plants and algae

Eukaryotic cells pass through a "cycle," progressing from a newly formed cell to a cell that is dividing to produce the next generation of progeny cells. Prior to division, the cell is in an "interphase"; during division, the cell is in a "division phase." During interphase, the nuclear DNA is organized in a dispersed network of **chromatin**, which is a complex consisting of nucleic acid and basic proteins. Immediately prior to and during division, the chromatin condenses to a series of discrete, compact structures called chromo-

chromatin complex of DNA, histones, and other proteins, making up chromosomes



An achievement celebrated in international headlines, Dr. J. Craig Venter of Celera Genomics published the human genome map, pictured here, in 2000. A genome is the entirety of an organism's DNA, and can be written as a long sequence of nucleotides (A, T, C, and G).

somes. Thus, the physical organization of the genome varies from interphase to division phase. Finally, viruses (which are noncellular, parasitic "life forms") have genomes of double-stranded DNA, single-stranded DNA, double-stranded RNA, or single-stranded RNA.

Eukaryotes

In sexually reproducing eukaryotes, progeny organisms receive a portion of their genetic information from each parent, receiving half the information from each. These parental contributions are designated **haploid** complements. The haploid complement can be represented as a "C value," which expresses the haploid complement as an amount of DNA measured in **base pairs**. Alternatively, the haploid complement can be expressed as the number of chromosomes contributed by each parent: This number of chromosomes is characteristic of each species. Finally, the haploid complement can be expressed as the number of genes on the haploid set of chromosomes.

haploid possessing only one copy of each chromosome

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

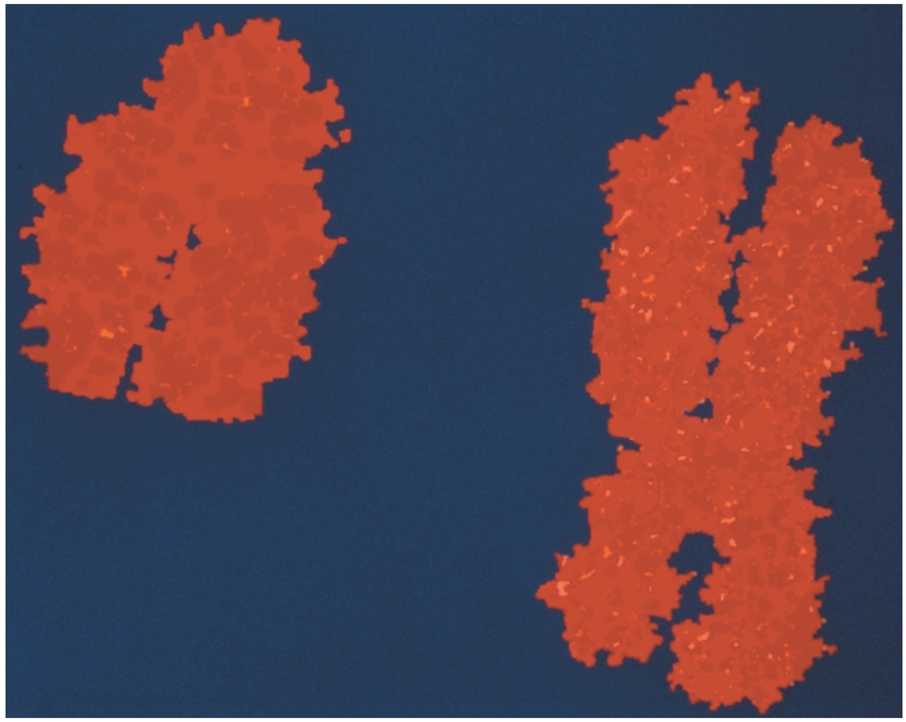
Chromosome Number

Each species has a characteristic number of chromosomes. For species with genetically determined sexes, the haploid set is composed of **autosomes** plus a sex chromosome. *Homo sapiens*, for example, have 22 autosomes plus an X chromosome or Y chromosome. The haploid DNA content of chimpanzees is nearly identical, but is organized into 23 autosomes plus a sex chromosome.

autosomes chromosomes that are not sex determining (not X or Y)

The record for minimum number of chromosomes belongs to a subspecies of the ant, *Myrmecia pilosula*. The females have a single pair of chromosomes, while males have only a single chromosome. Like some other members of the insect class, these ants reproduce by a process called haplodiploidy, in which diploid fertilized eggs develop into females, while haploid unfertilized eggs develop into males.

The record for maximum number of chromosomes is found in the plant kingdom, due to a condition known as polyploidy. In polyploidy, many extra sets of chromosomes beyond the normal diploid number may accumulate



Human Y and X chromosomes. These determine human gender.

over time. Cultivars of wheat exist with diploid numbers of chromosomes equaling 14, 28, or 42 (multiples of the haploid number, which is 7). Polyploids exist for many cultivated plants, including potatoes, strawberries, and cotton, as well as in wild plants such as dandelions. Polyploidy has led to striking numbers, and the known record is held by the fern *Ophioglossum reticulatum*, which has approximately 630 pairs.

Genome Size or C Value

The C value is the amount of DNA in a haploid complement. Currently, the amount is reported as the total number of base pairs. Generally, more complex organisms have more DNA. For example, the haploid complement of *Homo sapiens* DNA contains between 3.12 and 3.2 gigabases (the prefix "giga" denotes billions), while the haploid complement of yeast (*Saccharomyces cerevisiae*) DNA contains 12,057,500 base pairs.

Unexpected genomic sizes occur, however, in a condition called the C value paradox. Two closely related species can have widely divergent amounts of DNA. For example, *Paramecium caudatum* has a C value of 8,600,000 kilobases (where the prefix "kilo" denotes thousands) while its near relative *P. aurelia* has a C value of just 190,000 kilobases. Another paradoxical circumstance occurs when a simpler organism has a C value higher than a more complex organism. For example, *Amphiuma means* (a newt) and *Amoeba dubia* (an amoeba) have, respectively, C values that are 26 and 209 times the C value of humans.

Number of Nuclear Genes, "Gene Density," and Intergenic Sequences

An important trend in genome evolution has been the accumulation, both within the genes (intragenic) and between genes (intergenic), of DNA that does not code for any gene products. *Homo sapiens* have between 31,000 and 70,000 genes; mice have 24,780; *Caenorhabditis elegans* (a roundworm) has more than 19,099; fruit flies have 13,601; and yeast approximately 6,000. A ratio of gene number to C value indicates that lower organisms have both smaller genes and lower numbers of nongene base pairs between adjacent genes. Higher eukaryotes have a larger number of intragenic inserts (introns), greater intergenic distances, and more abundant repeated sequences.

In higher eukaryotes, only a small portion of the genome is organized into genes. For example, in humans less than 2 percent of the genome specifies protein products. Another portion (about 20 percent in humans) is present as gene fragments, pseudogenes (sequences that resemble genes but are not expressed as proteins), and surrounding stretches of nucleotides. The vast majority of nucleotides (approximately 75 percent in humans) constitute extragenic sequences. Two forms of extragenic sequences are prominent: unique sequences and repetitive sequences.

For repetitive sequences, two types of organization occur: short tandem repeats (called satellite sequences) and widely distributed, interspersed repeats. Satellites are recurrent short sequences present in essential chromosomal structures such as **centromeres** and **telomeres**. Interspersed repeats are generated from transposons, which are nucleotide sequences that can replicate themselves and become distributed throughout the genome. An example of interspersed repeats that occurs in humans is a sequence of a few hundred nucleotides called Alu, which occurs approximately a million times. In higher plants, satellites and interspersed sequences constitute the bulk of the genome.

centromeres regions of the chromosome linking chromatids

telomeres chromosome tips

Ploidy

Ploidy reflects the reproductive mechanisms of an organism. Animals commonly have both a maternal and a paternal parent. Through meiosis, the former forms a haploid **gamete** called an ovum (or egg); the latter forms a haploid gamete called a sperm. During fertilization, the egg and sperm unite to form a **diploid** zygote that matures to an adult organism. Thus, the genome of adult animals is diploid, while the genome of their gametes is haploid.

gamete reproductive cell, such as sperm or egg

diploid possessing pairs of chromosomes, one member of each pair derived from each parent

Plants exhibit an alternation of generations; sporophytes (the mature, visible plant) are diploid; through meiosis, they produce spores that germinate into gametophytes; the gametophytes are haploid and produce gametes that fuse to reestablish the diploid state. Fungi also exhibit an alternation of generations. They commonly exist as **multinucleate** tubes of cytoplasm called hyphae. The individual nuclei are most often haploid (though may be diploid in the lower fungi).

multinucleate having many nuclei within a single cell membrane

Hyphae of different members of a fungal species sometimes fuse; in this circumstance (called heterokaryosis) the genome becomes the sum of the two (dikaryotic) haploid complements. Unicellular protistan organisms, a group that includes protozoans and most algae, exhibit many variations. For example, the ciliates (such as paramecia) have diploid micronuclei and polyploid macronuclei; the former are the basis of inheritance; the latter establish the genetic character of an existing organism.

Mitochondrial and Chloroplast Genomes

Two cytoplasmic organelles responsible for the production of energy are the mitochondria (present in nearly all eukaryotic cells) and chloroplasts (present only in photosynthetic organisms). Both contain small, circular DNA molecules that constitute the nonnuclear portion of a eukaryotic genome. These organelles are descended from formerly free-living bacteria that took up residence in the first eukaryotes.

The human mitochondrial genome contains 16,569 base pairs specifying 13 protein products and 24 RNA products. In both lower eukaryotes and especially plants, larger mitochondrial genomes are present. In extreme cases, mitochondrial genomes may be several hundred thousand or millions of base pairs. Chloroplast genomes contain between 100 and 200 kilobases. It is thought that each was once larger, but over time their genes have been moved to the nucleus.

Prokaryotes

Prokaryotic genomes are composed of a chromosome plus various accessory elements. The former is most commonly a circular double-stranded DNA molecule but may be a linear molecule in some major groups, such as *Streptomyces* and *Borrelia* (the causative agent of Lyme disease). Accessory elements most prominently include plasmids (commonly circular but linear in *Actinomycetes* and some *Proteobacteria*) as well as insertion sequence (IS) elements, transposons, and prophages (derived from viruses). Other variations in chromosomal geometry exist: multiple circular chromosomes are found in some organisms; combinations of circular and linear chromosomes occur in others; and, in the extreme (observed in *Streptomyces*), circular and linear chromosomes can convert between those two topologies.

The smallest bacterial chromosome, with only 580 kilobase pairs (kbp) occurs in *Mycoplasma genitalium*, and the largest, with 9,200 kbp, occurs in *Myxococcus xanthus*. Representative sizes cluster between 2,000 and 5,000 kbp (e.g., *Escherichia coli* MG1655 has 4,649,221 bp). A typical bacterial gene contains approximately a thousand base pairs. *M. genitalium* has approximately 470 genes, while *M. xanthus* has more than 10,000, and *E. coli* has approximately 4,288.

By 2002 the nucleotide sequences of more than seventy-five prokaryotic chromosomes had been mapped. One goal of these sequencing projects is gene annotation: establishing the location, function, and **allelic variation** for each gene. In *E. coli* MG1655, for example, the positions of the 4,288 protein-coding genes have been identified; the average distance between genes is 118 base pairs; and the noncoding sequences (some of which may function as regulatory sites) constitute less than 11 percent of the genome. The function of approximately 40 percent of the genes, however, remains unknown. Notably, the chromosomal size and gene content of another isolate of *E. coli*, the pathogenic H157:O7 strain, are quite different. The H157:O7 chromosome is 20 percent larger, while MG1655 and H157:O7 share 4.1 million base pairs (mbp) in common. H157:O7 has 1.34 mbp that are not found in MG1655 and MG1655 has 0.53 mbp that are not found in H157:O7.

allelic variation presence of different gene forms (alleles) in a population

The genomes of closely related prokaryotes often have different organizations. These differences arise from rearrangements (such as inversions) between repeated elements, IS elements, and transposons and from the "horizontal transfer" of nucleotide sequences between cells. The latter phenomenon is mediated most commonly by conjugative plasmids, which are nonessential, autonomous accessory genetic elements that can acquire genes (such as antibiotic resistance genes) and then move them from a donor organism to a recipient. The dynamic character of genomic organization in prokaryotes is often designated as "genomic plasticity."

A series of repeated elements exist in the chromosomes of prokaryotes. In some instances the repeats are redundant copies of essential, long nucleotide sequences, as is seen in ribosomal RNA loci. Other repeats are small and have known functions (as in the Chi sequences in *E. coli* that facilitate genetic crossing over) or unknown functions (as in the REP [repeated extragenic palindromic] sequences in *E. coli*).

Viruses

Viral genomes are composed of single-stranded or double-stranded DNA or RNA. Single-stranded RNAs are either positive (capable of being immediately translated into protein) or negative. Double-stranded RNA genomes are most often segmented, with each segment being a single gene, while the other genomes are single circular or linear molecules. The *Retroviridae* have single-stranded RNA genomes that are converted by an enzyme (reverse transcriptase) into double-stranded DNA that becomes incorporated into the **genome** of the host.

genome the total genetic material in a cell or organism

The smallest known virus, containing 5,386 bases, is a member of the *Microviridae*, which infects bacteria and is designated fX174. The largest viral genomes occur in *Poxviridae*, which can possess as many as 309 kbp.

Viruses are extraordinarily efficient in using the coding capacity of their genomes. The virus known as fX174 contains ten genes, and the end of one gene commonly overlaps with the beginning of the following gene. In addition, two smaller genes are nested within larger genes (this compaction being achieved by having the two genes expressed in alternate "reading frames"). As a consequence of this efficiency, only 36 bases are not translated into an amino acid sequence. At the opposite extreme, the various pox viruses share more than 100 similar genes and may have an equal number of unique genes. SEE ALSO ARCHAEA; CELL, EUKARYOTIC; CELL CYCLE; CONJUGATION; EUBACTERIA; EVOLUTION OF GENES; GENE; GENOMICS; HUMAN GENOME PROJECT; POLYMORPHISMS; POLYPLOIDY; READING FRAME; REVERSE TRANSCRIPTASE; TRANSPOSABLE GENETIC ELEMENTS; VIRUS.

Steven Krawiec

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Genomic Medicine

A quote commonly heard these days is that in the history of medicine, the greatest advancements in the treatment of patients have occurred within the past 50 years. But what if a doctor could prevent a disease from occurring, treating the cause rather than the symptoms? We all agree this would be wonderful, but how could a doctor predict a patient's medical future? This dream is now within the realm of possibility. In fact, the greatest change in the history of medicine since the discovery of antibiotics is anticipated to occur over the next several generations. It is the movement of medicine from a discipline that reacts to and treats a disease to one that is focused on preventing disease from occurring.

This feat will be accomplished using genetic and genomic information gained from the Human Genome Project to tailor treatments and medicine to each individual patient. The individual's risk of getting a disease, or even infections, is almost always at least partly due to the combination of genes he or she was born with. Therefore, by using genetic information doctors will be able to predict, with different degrees of certainty, what diseases their patients are at risk of developing. They will then be able to use this information to reduce the chance of the disease from occurring in an individual, or prevent it entirely. This new approach of using genetics in the regular practice of medicine has been given the name "genomic medicine."

The Importance of SNPs

Genomic medicine will be applied to patients through several mechanisms. The primary tool will be the detection of single nucleotide **polymorphisms** (SNPs). These are small variations in DNA sequences that are found in every person. These normal polymorphisms are very frequent, occurring in approximately one out of every 1,000 base pairs. The majority of these SNPs occurs in regions of the genome that do not code for proteins, and usually do not contribute to a person's disease susceptibility. However, they may serve as "markers" for a disease, lying close to an important susceptibility gene. When they occur directly in a protein-coding area of a gene (**exon**), they can cause the protein variation that helps make each of us unique. This variation in proteins is one of the primary reasons that each of us differs in our risk for disease, infections, and drug tolerance.

polymorphisms DNA sequence variant

exon coding region of genes

mutations changes in DNA sequences

It is important to realize that these SNPs are not **mutations**. The term "mutation" suggests a nucleotide change that prevents a protein from performing its normal function. These mutation-caused changes are rare, occurring in less than 1 percent of the population. This type of genetic mutation is almost always severe enough that its presence alone is enough to cause the disease. In contrast, SNPs are much more common, certainly occurring in more than 1 percent of the population. Their presence may make it easier for a person to get a specific disease, but they do not cause the disease by themselves. Instead, disease can occur only in the presence of the correct environment, other gene combinations, drugs, and other such factors. This is very important, for it suggests that by knowing that an individual is at risk for a disease, a doctor can take actions to prevent it. These actions may be as simple as a dietary or lifestyle change, or taking a medicine or vaccine.

Doctors will be able to use this same identification of SNPs to screen patients prior to prescribing drugs. Drug side effects are usually due to a

This DNA chip created by Toshiba was designed to store an individual's genetic profile. The electronics company expects to release this electrochemical technology in 2003.

specific, uncommon polymorphism in a gene that produces a protein that interacts with the drug. These gene interactions may not have anything to do with how the drug helps the patient. More likely, the gene is involved in how the body metabolizes the drug. This field of studying genes and their interaction with drugs is called pharmacogenetics. And pharmacogenetics does not just study drug side effects. Every doctor knows that some drugs work better for some patients than others. This difference is also likely due to normal polymorphisms, or different genetic forms of the disease.

The Future of Medicine

The idea of preventive medicine is not new, but until the completion of the Human Genome Project medicine did not have a way of accomplishing it for common medical problems. For example, for years almost every baby born in the United States has had its urine or blood screened early in life for phenylketonuria (PKU), a rare genetic disease in which affected individuals cannot metabolize the amino acid phenyalanine, a common amino acid in our food. Untreated, PKU patients develop severe mental and psychomotor retardation. However, if a child is identified early in life to have the PKU mutation, the disease can be prevented by placing the child on a special diet that lacks phenylalanine.

Genomic medicine technique will not look at just one gene or protein in an individual. It will also look at the interaction of thousands of genes at once, using DNA chips or microarrays. These microarrays are already being used in cancer chemotherapy to predict which drugs will work best on each patient's specific tumor.

Therefore, in the future an individual will be able to go to the doctor for a regular checkup and give a small sample of blood, or maybe even just let the doctor take a mouth swab. This will provide a DNA sample that would then be placed on a genetic "microchip" or other device and quickly be genotyped to give a genetic profile of the patient. The doctor will use the information to tailor the medical treatment for that patient. Lifestyle changes or medicine will be suggested to prevent the occurrence of diseases to which the patient is genetically susceptible, or at least to reduce the risk or severity of such diseases. If medication must be prescribed, the doctor will also use this genetic profile before choosing the medicine, to make sure that "the right medicine for the right patient" is chosen: one that works and will not harm the patient or cause side effects.

These changes in medicine are likely to take place over the first half of the twenty-first century. They will be exciting, but they will require that both patient and doctor have a strong understanding of genetics, the most powerful future tool of medicine. SEE ALSO DNA MICROARRAYS; GENETIC TESTING; PHARMACOGENETICS AND PHARMACOGENOMICS; POLYMORPHISMS.

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Genomics

Genomics is a recent scientific discipline that strives to define and characterize the complete genetic makeup of an organism. Its primary approaches are to determine the entire sequence and structure of an organism's DNA (its **genome**) and then to determine how that DNA is arranged into genes. This second goal is accomplished by determining the structure and relative abundance of all messenger RNAs (mRNAs), the middlemen in genetics that encode individual proteins.

genome the total genetic material in a cell or organism

From Microorganisms to Human DNA

For many years, genomics has been focused on microorganisms, which have relatively small genomes. However, more recently the field has been energized by the advent of more industrialized, higher-throughput sequencing technologies. By 2001 more than seventy organisms had been completely sequenced, and a working draft of the human genome had been produced. Vigorous efforts have now been initiated to map the mouse genome, and one company already claims to have completed the sequence. From the description of the structure of the genetic material by James Watson and Francis Crick in 1953, it will have taken only about fifty years to determine the complete genetic codes of humans and most of the model organisms that are important in biological research.

Latin Name	Common Name	Genome Size
Eukaryotes (haploid genome)		
Oryza sativa	Rice	420,000 Kb
Homo sapiens	Human	3,200,000 Kb
Arabidopsis thaliana	Mustard cress	115,428 Kb
Drosophila melanogaster	Fruit fly	137,000 Kb
Caenorhabditis elegans	Roundworm	97,000 Kb
Saccharomyces cerevisiae	Yeast	12,069 Kb
Eubacteria		
Haemophilus influenzae	–	1,830 Kb
Escherichia coli	Human colon bacterium	4,639 Kb
Helicobacter pylori	Stomach ulcer bacterium	1,667 Kb
Mycobacterium	Tuberculosis	4,411 Kb
Yersinia pestis	Plague	4,653 Kb
Archaea		
Halobacterium	Salt-tolerant archaean	2,014 Kb
Methanobacterium thermoautotrophicum	Methane-producing archaean	1,751 Kb

Kb=one thousand base pairs

Size comparison of selected completed genomes. Most of these organisms are of economic, medical, or scientific importance.

Of what value is the knowledge of these genomes? How are they being used within the scientific community? The first fully sequenced genomes included the fruit fly, a worm, and a number of bacteria and yeast. One of the first analyses performed was to simply compare the sequences between organisms, in order to identify what is shared in common and what is different. This allows the very specific comparison of organisms that will enable the refining of phylogenic relationships. This kind of information is also very valuable for asking questions about how organisms have evolved, how they adapt to different circumstances, and what gene products contribute to their survival in various environmental conditions.

Applications

Genomics has brought us to the threshold of a new era in controlling infectious diseases. These studies will likely lead to the development of new disease prevention and treatment strategies for plants, animals, and humans alike. For instance, understanding pathogen genes, their expression, and their interaction will lead to new antibiotics, antiviral agents, and "designer" immunizations. These new DNA-based immunizations are by-products of genomic research and will undoubtedly eventually replace the traditional vaccines made from whole, inactivated microorganisms. This is highly relevant to domesticated animals, where viruses still kill billions of dollars worth of livestock every year.

Understanding the genomes of plants and animals has additional benefits. Gene mapping should allow us to understand the basis for disease resistance, disease susceptibility, weight gain, and determinants of nutritional value. The use of genomic information provides the opportunity to select optimal environments for the healthy growth of plants and animals, to develop disease-resistant strains, and to achieve improved nutritional value such as with the "golden" rice. Success in these species may well provide important insights needed to improve the health of humans.

The Human Genome Project and Future Research

The Human Genome Project reached a major milestone in 2001, with two separate publications of working drafts of the human genome. Although much knowledge has been generated, the sequence is not complete. Neither the actual number of genes nor all their structures have been determined. However, several major lessons have been learned. First the number of genes is estimated to be between 30,000 and 70,000, fewer than previously thought. In addition, it is clear that a very large proportion of our genes are highly similar to those in other organisms, such as the fruit fly and the microscopic worm, *C. elegans*. The observation that we can build humans with between 30,000 and 70,000 genes and a fruit fly with 15,000 genes suggests that we owe much of the complexity of humans to the fine regulation of genes and not their absolute number.

Genomics has also forced biologists to begin to look at the function of genes in an industrialized mode. This new field of functional genomics takes advantage of a number of new technologies. Since many fly and worm genes are so similar to human genes (homologs), these animals can be used as model systems to study gene function. In these model systems it is possible to mutate (or alter) the structure of every single gene, enabling researchers to determine each gene's function and how several of the genes interact in complex metabolic pathways. Similar efforts using systematic gene mutations are also underway to create DNA "libraries" of two vertebrates, mice and zebrafish, whose genes are surprisingly similar to humans. Once these genomes are fully sequenced and characterized, it will be possible to create animals with disorders that are more precisely like those of humans, allowing for a better understanding of complex diseases and determination of novel and effective therapies.

Genomics allows for the comparison of sequences between individuals, too. These studies can be used as a basis for the understanding and diagnosis of disease, especially of the complex disorders not governed by single genes. Knowledge of the entire human sequence is also the basis of the fields of pharmacogenetics and pharmacogenomics. Pharmacogenomics seeks a broader understanding of how genes influence drug response and toxicity, and the discovery of new disease pathways that can be targeted with tailor-made drugs. Pharmacogenetics is the study of the genetic factors involved in the differential response between patients to the same medicine. Polymorphisms, **nucleotide** changes that occur in more than 1 percent of the population, are the basis for our individuality but also account for our differential susceptibility to disease and the variable outcome of treatments. Through a variety of research efforts, more than one million polymorphisms have been identified in the human genome. The study of these variants, that occur once every 500 to 1,000 nucleotides in the human genome, should enable pharmacogenetics to define the optimal treatment regimens for subsets of the population, allowing a wider range of patients to be treated and more effective outcomes to be produced with any given drug. SEE ALSO Agricultural Biotechnology; DNA Libraries; Genomic Medicine; Genomics Industry; High-throughput Screening; Human Genome Project; Model Organisms; Pharmacogenetics and Pharmacogenomics.

nucleotide the building block of RNA or DNA

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Genomics Industry

Fundamental to the myriad of genomic research efforts in operation around the world is the mapping and sequencing of whole **genomes**. The entire genomes of more than seventy organisms had been completed by early 2002, including the working drafts of the human genome, first published in 2001. The successful completion of the sequencing of these genomes was made possible in part by companies developing and utilizing new technologies and processes to increase the speed and accuracy of mapping and sequencing. This effort has also spawned entire new fields that aim, in various ways, to capitalize on genomics and identify disease-causing genes and new therapeutic strategies.

genomes the total genetic material in cells or organisms

The advent of the global Human Genome Project in 1989 provided significant, additional incentive for the development of a variety of new genomics-based companies. These companies can be loosely divided into seven major types: large-scale sequencing companies, gene mining companies, functional genomics companies, population-based genomics companies, **bioinformatics** companies, established pharmaceutical companies, and new biopharmaceutical companies. These are general distinctions and are not absolute, since many companies are blending and using a variety of these technologies with overlapping applications.

bioinformatics use of information technology to analyze biological data

Large-Scale Sequencing Companies

Early stages of the human genome project involved the physical mapping and subsequent sequencing of genes and intervening segments. A number of companies were founded specifically to identify genes, sequence them, and determine their function as a means to lead to new diagnostics and pharmaceuticals. These companies include Incyte, Human Genome Sciences, and Celera Genomics. This group of companies has contributed to public and private databases millions of bases of sequence data not only for the human genome, but for many microorganisms as well.

Gene Mining Companies

Now that the sequence data is available and placed in the public domain, companies have been created to "mine" the data, that is, to analyze the genomic sequences to identify genes, their function, and their relationships to health and disease processes. Companies pioneering in this area included Sequana and Millennium Pharmaceuticals (although Sequana did not survive as an independent company). Once genes were identified as relevant to specific disease processes, some of the companies in this sector focused on discovering and developing small molecules, **antibodies**, proteins, or a combination of the three, in search of drugs that will target the consequences of the defective gene or dysregulated pathways.

antibodies immune-system proteins that bind to foreign molecules

Functional Genomics Companies

The challenge of the postgenomics era has been to identify those genes that are of clinical value for drug development. Inherent to that is a basic understanding of the function of all genes. Using industrialized biology approaches, these companies develop technologies for their own use or for sale to other companies. These new technologies are designed to pinpoint specific genes associated with disease and to determine causal relationships with **pathology**. Companies specializing in this area include Affymetrix, Excelesis, and Lexicon. In addition, most major pharmaceutical companies have established functional genomics departments.

pathology disease process

Population-Based Genomics Companies

In order to determine which genes are relevant and underlie complex human diseases such as diabetes mellitus, Alzheimer's disease, or hypertension, companies have been created to collect patient materials. These firms collect relevant clinical information and DNA on people suffering from defined disease as well as people who have no known disorders. They then look at DNA sequence variations in an attempt to identify the genetic factors that may predispose an individual to develop these types of disorders, or factors that directly lead to the development of disease. One company working in this area is deCODE, which has access to the genetic and health data of the entire population of Iceland.

Bioinformatics Companies

In order to extract the maximum amount of information from genomic sequence data, a new discipline has emerged, called bioinformatics. Bioinformaticians are involved in the capture and interpretation of biological data. The need for bioinformatics support throughout the genomics industry is significant and, as a result, a number of companies have been started that provide these services. Companies engaged in this type of research include DoubleTwist, Genomica, and Spotfire. Because genomics has created terabytes (a trillion bytes) of data, virtually all genomics efforts now require a substantial investment in in-house bioinformatics.

Established Pharmaceutical Companies

The pharmaceutical industry has been working to integrate genetic and genomic information into the drug discovery process. Essentially all of the major pharmaceutical companies have such programs, including GlaxoSmithKline, Merck, and Novartis. From the beginning of compound synthesis through the identification of drug targets, genomics is changing the traditional drug development process. Understanding the influence of **polymorphisms** on drug target binding is critical to optimizing drug development. This is especially important now, because successes in combinatorial

polymorphisms occurring in several forms chemistry have given rise to a tremendous increase in the number compounds to be analyzed.

New Biopharmaceutical Companies

As a result of molecular biology and genomics, a number of new companies have been formed that focus on creating drugs from specific genes. As genes are identified and the function of the protein is learned, these companies manufacture the proteins made by specific genes for use in the treatment of human disease. This includes a variety of drugs made through recombinant DNA technology, such as insulin and growth hormone. These new drugs provide patients with a safer therapeutic alternative to other protein sources. Companies working in this sector include Genentech and Amgen. In addition, some companies focus on making protein therapeutics in novel ways. One such company is Genzyme Transgenics, which has genetically altered livestock to produce therapeutic protein in their milk. Other companies focus on producing artificial **antibodies** that are engineered to look just like normal human antibodies. These so called humanized antibodies, produced by companies such as Protein Design Laboratories, are already on the market for use in treating diseases such as cancer and arthritis.

antibodies immune system proteins that bind to foreign molecules

From Present to Future

Technological advances clearly drive innovation in the pharmaceutical and health-care industries. Advances in the speed and accuracy of DNA sequencing and the growing understanding of the genome have created hundreds of companies, with billions of dollars in assets, that are using this information to contribute to the creation of new diagnostic and therapeutic agents. This has been accomplished even before science has achieved full knowledge of the human DNA sequence or a complete understanding of the function of most human genes. It is expected that the next decade will provide the remaining human genomic information, which will enable a breathtaking array of new therapeutic and preventive options for the treatment of human disease. SEE ALSO Bioinformatics; Biotechnology; Combinatorial Chemistry; Genome; High-throughput Screening; Human Genome Project; Pharmacogenetics and Pharmacogenomics.

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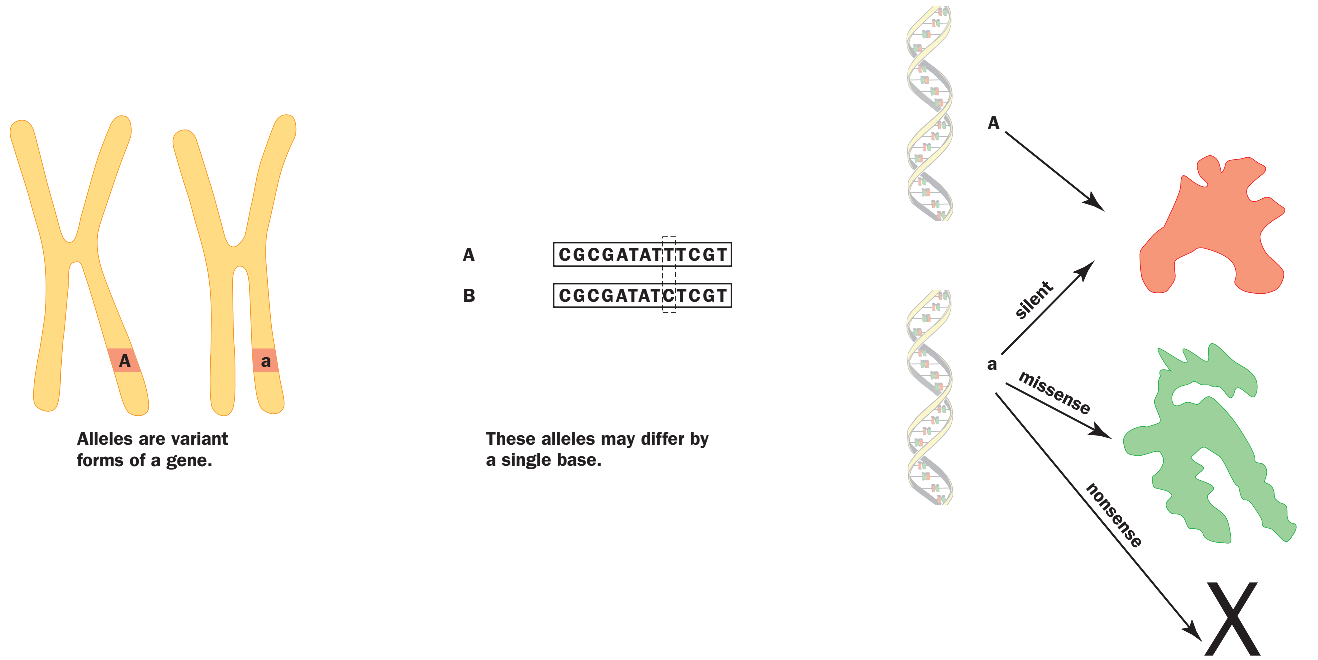
Genotype and Phenotype

An individual's genotype is the composition, in the individual's **genome**, of a specific region of DNA that varies within a population. (The genome of the individual is the total collection of the DNA in a cell's chromosomes. It includes all of the individual's genes, as well as the DNA sequences that lie between them.)

genome the total genetic material in a cell or organism



Alleles are variant forms of a gene.

These alleles may differ by a single base.

The sequence change may have no effect on the protein, may cause a change in structure or function, or may prevent it from being formed.

Alleles differ in their DNA sequence, and may lead to production of alternative forms of the protein they encode.

A genotype could represent a single DNA nucleotide, at a specific location on a chromosome. It could also be a sequence repeated multiple times, a large duplication, or a deletion. Most variation in genotypes does not cause any difference in the proteins being produced by the cell, because genes, which code for proteins, occupy only about 2 percent of the total genome. However, when a specific genotype does affect the composition or expression of a protein, disease or changes in physical appearance can result. The physical effect of a particular genotype is known as its phenotype, or trait.

Alleles, Polymorphisms, and Mutations

An individual genotype is composed of two distinct parts, the inherited sequences from the maternal and paternal genomes. Therefore, for every genotype, there are two copies of the sequence. For genes, we refer to these individual sequence copies on each chromosome that make up a genotype as the two alleles. Alleles may be identical or different, and various combinations of alleles can create a range of phenotypes.

polymorphisms occurring in several forms

mutations changes in DNA sequences

A DNA change within a gene may or may not alter the protein that is encoded by the gene. If the change contributes to normal genetic variability and is found in more than 1 percent of the population being studied, the variation is called a **polymorphism**. Variations in a gene that modify, seriously disrupt, or prevent the functioning the encoded protein are called **mutations**. The effect of a mutation can range from harmless to harmful depending on the type of mutation.

One type of sequence variation is the substitution of one **nucleotide** for another, which can thus affect the three base-pair unit (codon) that codes for a specific amino acid. In some cases, where the substitution does not change the amino acid, the result is a neutral or silent mutation. A substitution that alters the coding sequence, and thus substitutes a different amino acid, is called a missense mutation. This type of mutation can have a beneficial effect whereby there is a positive gain of function for the protein, it can also have a drastically negative effect and result in loss of function or the gain of a new function that is deleterious to the organism. By far the most severe change is a nucleotide substitution that creates a stop codon where one should not be. This phenomenon, called a nonsense mutation, will prevent the full formation of the protein. A nonsense mutation can completely abolish the function of the protein, which can be lethal to the organism if the protein is essential for sustaining a key biological process.

nucleotide the building block of RNA or DNA

Dominant, Codominant, and Recessive Genes

Differences in phenotype can result from differences in genotype. Two individuals may have different bases at a particular location in the coding region of a gene. While this will result in different codons in that part of the sequence, the two different codons may code for the same amino acid, which will in turn result in the same protein. Thus it can be said that the two individuals have different genotypes in their DNA, but that their phenotypes are identical. This example illustrates the first factor one must look for in predicting phenotypes: how the genotype affects the translation of DNA.

The second factor is the inheritance pattern of the genotype. If each parent transmits an identical gene sequence, A, to a child, then the genotype of the offspring will be AA. This is called a **homozygous** genotype. However, if the child inherits a different allele, a, from one of the parents, then its resulting genotype, Aa, contains two different sequences. Such genotypes are called **heterozygous**.

homozygous containing two identical copies of a particular gene

heterozygous characterized by possession of two different forms (alleles) of a particular gene

An Aa genotype can result in the same phenotype as either an AA or aa genotype, if one of the alleles acts in a dominant fashion. If the A allele is dominant over the a allele, then the phenotype of a heterozygous (Aa) individual will be the same as the phenotype of a homozygous dominant (AA) individual.

Huntington's disease, a nervous system disorder, follows a dominant inheritance pattern. The presence of one mutated Huntington gene will result in the conditions of the disease. Even if a patient has one normal Huntington gene on another chromosome, one mutated copy is enough to produce Huntington's disease. Thus, the Huntington allele is said to be dominant over the normal allele.

Another type of inheritance pattern is called recessive. Here, two copies of the mutant allele, a, must be inherited (resulting in genotype aa) to cause a change in phenotype. This is exemplified by cystic fibrosis, a recessive disorder marked by digestive and respiratory problems. In this case a heterozygous individual (genotype Aa) who carries one copy of the faulty gene will not display clinical symptoms, because the normal gene is dominant over the *CFTR* gene. The normal gene is able to express the protein that **epithelial cells** need for transporting chloride. If two parents are

epithelial cells one of four tissue types found in the body, characterized by thin sheets and usually serving a protective or secretory function heterozygous for the mutated cystic fibrosis gene, there is a 25 percent chance that their child will inherit a mutated copy from each parent and will have the disease. Heterozygous carriers, who live normal lives, pass on the mutant gene to half of their children, enabling it to stay within the population for generations and to persist at relatively high frequencies.

Although it is convenient to illustrate inheritance concepts by talking about diseases, if we consider the diversity of phenotypes expressed by all organisms in the living world it is obvious that not all variation is bad and most genetic changes do not lead to disease.

Multiple Alleles and Pleiotropy

loci sites on a chromosome (singular, locus)

Some genetic **loci** are multiallelic, having more than one allele that will manifest in a variety of phenotypes. In most mammal species, for example, the immune response genes of the major histocompatibility complex are extremely polymorphic—meaning that there are many different alleles at each gene locus. The combination of alleles in each individual may result in either susceptibility or resistance to specific disease-causing agents.

The human blood group system is another example of multiple alleles resulting in many different phenotypes. The genes that determine ABO blood type encode enzymes that add particular sugar groups to proteins in blood cells. A person's specific blood type is due to the presence or absence of A and B sugar-protein complexes on the surface of red blood cells.

There are three alleles involved, A, B, and O, and six possible genotypes: AA, BB, OO, AB, AO, and BO. The various genotypes result in four different phenotypes or blood types: A, B, O, and AB. Individuals have blood type A if their genotypes are AA or AO. Individuals have blood type B if their genotypes are BB or BO. Individuals have blood type O if their genotype is OO, and they have blood type AB if their genotype is AB.

Many of these examples describe the concept of genotype and phenotype in terms of proteins or diseases that have been thoroughly analyzed by scientists. However, genotypic differences are abundant in nature and are evident in the most extraordinary ways.

A small mutation in a viral gene may make an otherwise harmless strain of the influenza virus capable of causing disease or even death. Other genetic variations in mammals, including humans, may influence aggression or other social interactions.

Some genes affect more than one unrelated characteristic. These genes are said to be pleiotropic. One gene, for example, produces melanin, which is responsible for skin pigmentation and is also involved in nerve pathways. If there is a certain mutation in the melanin gene, no melanin will be produced. In humans, this condition, called albinism, causes white skin and, usually, vision problems. In domestic cats, it causes a white-hair, blue-eye phenotype along with hearing loss.

In the past, scientists thought that pleiotropic genes were unusual. However, the Human Genome Project has shown that humans have only about one-third of the number of genes that was predicted. It is now believed that most genes are pleiotropic, serving many functions. SEE ALSO ALTERNATIVE SPLICING; BLOOD TYPE; DISEASE, GENETICS OF; GENE; GENETIC CODE;

IMMUNE SYSTEM GENETICS; INHERITANCE PATTERNS; MUTATION; PLEIOTROPY; POLYMORPHISMS.

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Growth Disorders

Growth, which usually refers to skeletal growth since it determines final adult height, is an extremely complex process. As such, it is susceptible to a wide range of genetic and physiologic disturbances. Indeed, growth is adversely affected by many if not most chronic diseases of childhood, through many different mechanisms.

Skeletal growth depends on hormonal signals for regulation. It also requires the production of adequate amounts of cartilage, because most bone forms within a model or template made from cartilage. Primary disorders of growth, that is, disorders in which growth is intrinsically affected, therefore fall into two major categories: disorders of the endocrine (hormone) system and disorders of the growing skeleton itself (skeletal dysplasias). Many of the former and most of the latter are genetic disorders.

Endocrine Disorders

Growth hormone (GH) is produced by the pituitary gland at the base of the brain and is a major regulator of growth. Deficiency of the hormone is the prototype of the inherited endocrine disorders of growth. Although normal in size at birth, infants with GH deficiency exhibit severe postnatal growth deficiency while maintaining normal body proportions. If untreated, children typically have a "baby-doll" facial appearance and a high-pitched voice that persists after puberty.

Isolated GH deficiency most often results from deletion of all or part of the GH gene. Humans carry two copies of the GH gene, but having just one good copy is usually sufficient to prevent GH deficiency. Thus, this disorder is inherited as an autosomal recessive trait. Rarely, **point mutations** of this gene can lead to a dominantly inherited form of GH deficiency, in which the product of the mutant GH **allele** is thought to interact with and prevent secretion of the product of the normal GH allele.

point mutations gains, losses, or changes of one to several nucleotides in DNA

allele a particular form of a gene

transcription factors proteins that increase the rate of gene transcription

GH deficiency also results from mutations of genes that encode **transcription factors**, such as PIT1, PROP1, and POU2F1, which are necessary for development of the pituitary gland and of the cells that produce pituitary hormones. Patients usually have small pituitary glands and exhibit deficiencies of several pituitary hormones, including gonadotropins (FSH, LH), prolactin, and thyroid-stimulating hormone (TSH) in addition to GH. Multiple pituitary hormone deficiency of this type is inherited in an autosomal recessive fashion.

At their target cells, hormones exert their efforts by binding to receptors. The clinical manifestations of GH deficiency can also result from mutations of the GH receptor, in the autosomal recessive Laron syndrome. There are also a number of birth-defect syndromes in which hypopituitarism (reduced pituitary output) results in the abnormal development of craniofacial structures. Examples include anencephaly, holoprosencephaly, Palister-Hall syndrome, and some cases of severe cleft lip and cleft palate.

Deficiencies of other hormones relevant to growth and their receptors also occur on a genetic basis. For instance, thyroid hormone deficiency can be due to reduced TSH, as discussed above, but it can also result from loss-of-function mutations of enzymes that are involved in thyroid hormone biosynthesis. There are also several forms of thyroid hormone resistance due to mutations of thyroid hormone nuclear receptors. The biosynthetic defects are inherited as recessive traits, whereas thyroid resistance is usually inherited in a dominant fashion. Mental retardation, growth deficiency, and delayed skeletal development are the main clinical manifestations of thyroid hormone deficiency.

Skeletal Dysplasias

In contrast to endocrine growth disorders, the hallmark of the skeletal dysplasias ("-plasia" means "growth") is disproportionate short stature. In other words, the limbs are disproportionately shorter than the trunk or vice versa. These disorders result from mutations of genes whose products are required for normal skeletal development. In most cases they are involved in endochondral **ossification**, the process by which the skeleton grows through the production of the cartilage template that is converted into bone. The mutated genes encode cartilage and bone extracellular matrix proteins, growth factors, growth factor receptors, intracellular signaling molecules, transcription factors, and other molecules whose functions are needed for bone growth.

ossification bone formation

Growth Factor Receptor Mutations. The prototype of the skeletal dysplasias is achondroplasia, which is one of a graded series of dwarfing disorders that result from activating mutations of **fibroblast** growth factor receptor 3 (FGFR3). Achondroplasia is the most common form of dwarfism that is compatible with a normal life span, while thanatophoric dysplasia, which lies at the severe end of the spectrum of FGFR3 disorders, is the most common lethal dwarfing condition in humans. Both are characterized by the shortening of limbs, especially proximal limb bones, and a large head with a prominent forehead and hypoplasia (reduction of growth) of the middle face. The mildest disorder in this group is hypochondroplasia, in which patients exhibit mild short stature and few other features.

fibroblast undifferentiated cell normally giving rise to connective tissue cells

All of the disorders in this group result from **heterozygous** mutations of FGFR3. Except for the lethal thanatophoric dysplasia, they are inherited as autosomal dominant traits. The vast majority of mutations arise anew, during sperm formation (spermatogenesis), and especially in older fathers. FGFR3 is a very mutable (easily mutated) gene and there are certain extremely mutable regions within the gene where disease-causing mutations cluster.

heterozygous characterized by possession of two different forms (alleles) of a particular gene

There is a very strong correlation between clinical **phenotypes** and specific mutations. In fact, essentially all patients with classic features of achondroplasia have the same amino acid substitution in the receptor. The mutations that cause these disorders enhance the transduction of signals through FGFR3 receptors in **chondrocytes** in growing bones. This inhibits the proliferation of these cells that is necessary for linear growth to occur.

phenotypes observable characteristics of an organism

chondrocytes cells that form cartilage

A six-year-old girl with achondroplasia (right) stands beside her younger, not quite four-year-old brother. Features of achondroplasia evident in the girl include small stature, short arms and legs, relatively large head size, and subtle differences of facial features.

Cartilage Matrix Protein Mutations. Another major class of skeletal dysplasias result from mutations of genes that encode cartilage matrix proteins such as collagen types II, IX, X, and XI, and cartilage oligomeric matrix protein (COMP). The type II collagen mutations cause a spectrum of autosomal dominant disorders called spondyloepiphyseal dysplasias because they primarily affect the spine (spondylo) and the ends of growing limb bones (epiphyses). They range in severity from lethal before birth to extremely mild. In addition to dwarfism that affects the trunk more than the limbs, patients with these disorders develop precocious **osteoarthritis** of weight-bearing joints such as the hips and knees. Many patients have eye problems that reflect disturbances of type II collagen in the vitreous portion of the eye.

osteoarthritis a degenerative disease causing inflammation of the joints